

Fundamentals of Cyber Security and Digital Forensics

Module: I- Introduction of Cybersecurity

Priyanka Shingane

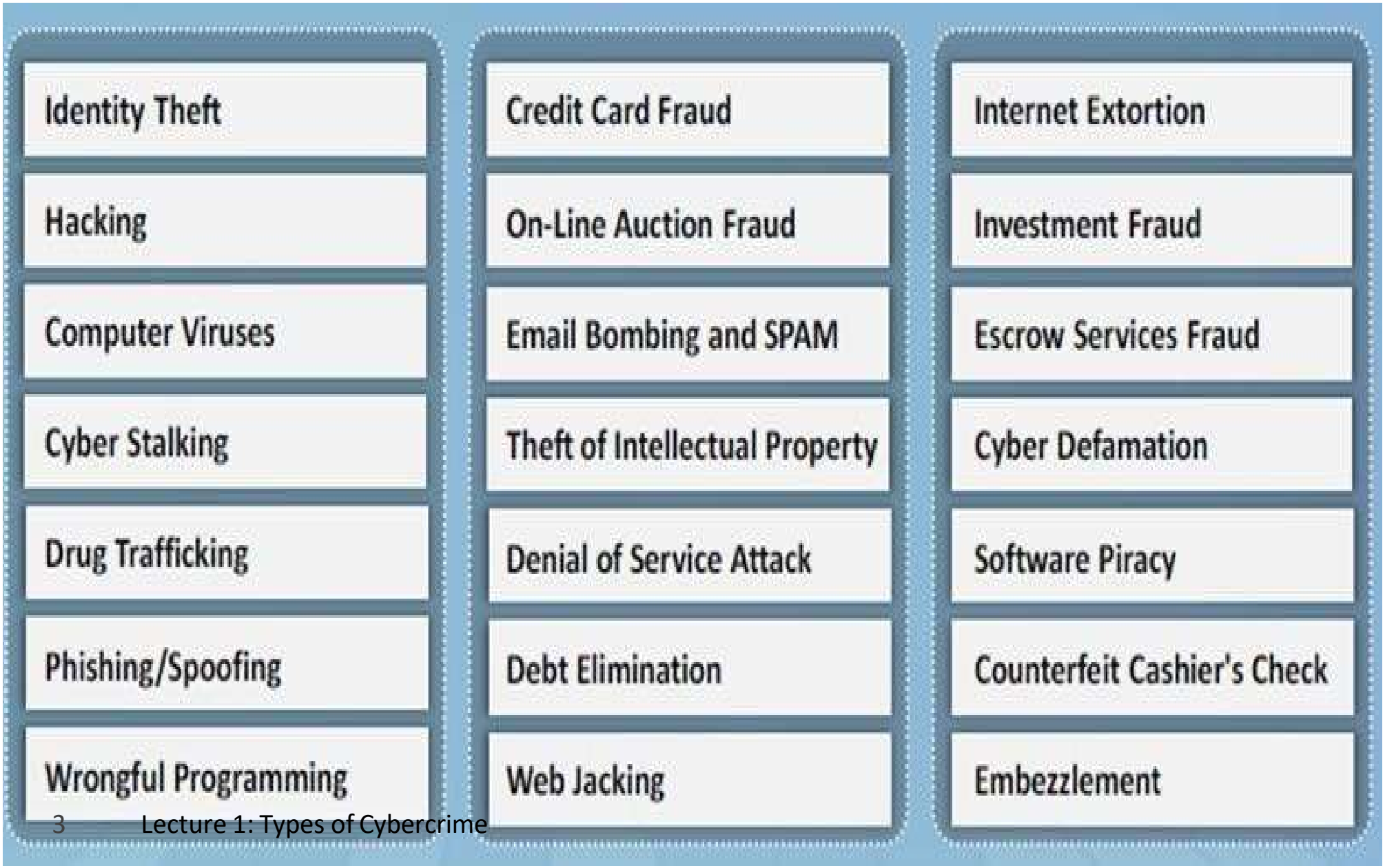
Module 1:

Lecture 2: Roles of Computer in Cybercrime, Categories of Cybercrime



Types of Cybercrime

https://www.youtube.com/watch?v=5k24We8pED8



Vulnerabilities and Threat

- A vulnerability is a weakness/flaw that can be exploited by cybercriminals to gain unauthorized access to a computer system.
- Threat is a process that magnifies the likelihood of a negative event, such as the exploit of a vulnerability.
- Risk is the potential for loss, damage or destruction of assets or data caused by a cyber threat.



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Vulnerabilities and Threat

	Vulnerability	Threat	Risk
Example 1	Terminated employee ID's are not removed from the system	Dialing into the company's network and accessing proprietary info	Unauthorized disclosure of sensitive business information
Example 2	Improper maintenance of fire fighting equipment	Fire	Loss of life, data and infrastructure

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Causes of Vulnerabilities

- Complex Systems – Complex systems increase the probability of misconfigurations, flaws, or unintended access.
- Familiarity – Attackers may be familiar with common code, operating systems, hardware, and software that lead to known vulnerabilities.
- Connectivity – Connected devices are more prone to have vulnerabilities.
- Poor Password Management – Weak and reused passwords can lead from one data breach to several.
- OS Flaws – Operating systems can have flaws too. Unsecured operating systems by default can give users full access and become a target for viruses and malware.

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Causes of Vulnerabilities

- Internet – The internet is full of spyware and adware that can be installed automatically on computers.
- Software Bugs – Programmers can sometimes accidentally, leave an exploitable bug in the software.
- Unchecked user input – If software or a website assumes that all input is safe, it may run unintended SQL injection.
- People – Social engineering is the biggest threat to the majority of organizations. So, humans can be one of the biggest causes of vulnerability.

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Types of Vulnerabilities

- System Misconfigurations:
- Out-of-date or Unpatched Software:
- Missing or Weak Authorization Credentials
- Malicious Insider Threats
- Missing or Poor Data Encryption
- Zero-day Vulnerabilities

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System Misconfigurations

- Network assets can cause system mistakes with incompatible security settings or restrictions.
- Networks are frequently searched for system errors and vulnerable spots by cyber criminals.
- Network misconfigurations are increasing as a result of the quick digital revolution.
- Working with knowledgeable security professionals is crucial when implementing new technology.
- Cybercriminals frequently search networks for vulnerabilities and misconfigurations in the system that can exploit.

Out-of-date or Unpatched Software

- Hackers frequently scour networks for vulnerable, unpatched systems that are prime targets, just as system configuration errors do.
- Attackers may use these unpatched vulnerabilities to steal confidential data, which is a huge threat to any organization.
- Establishing a patch management strategy that ensures all the most recent system updates are applied as soon as they are issued is crucial for reducing these types of threats.

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Missing or Weak Authorization Credentials

- Attackers frequently utilize brute force methods, such as guessing employee passwords, to gain access to systems and networks.
- Therefore, they must therefore train employees on cybersecurity best practices to prevent the easy exploitation of their login credentials.
- An endpoint system security will be a great addition to all laptop or desktop devices.

Malicious Insider Threats

- Employees with access to vital systems may occasionally share data that enables hackers to infiltrate the network, knowingly or unknowingly.
- Because all acts seem genuine, insider threats can be challenging to identify.
- Consider purchasing network access control tools and segmenting your network according to employee seniority and experience to counter these risks.

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Missing or Poor Data Encryption

- If a network has weak or nonexistent encryption, it will be simpler for attackers to intercept system communications and compromise them.
- Cyber adversaries can harvest crucial information and introduce misleading information onto a server when there is weak or unencrypted data.
- This may result in regulatory body fines and adversely jeopardize an organization's efforts to comply with cyber security regulations.

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Zero-day Vulnerabilities

- Zero-day vulnerabilities are specific software flaws that the attackers are aware of but that a company or user has not yet identified.
- Since the vulnerability has not yet been identified or reported by the system manufacturer, there are no known remedies or workarounds in these situations.
- These are particularly risky because there is no protection against them before an attack occurs.
- Exercising caution and checking systems for vulnerabilities is crucial to reducing the risk of zero-day attacks.

Internal Threats and Attacks

- An internal attack occurs when an individual or a group within an organization seeks to disrupt operations or exploit organizational assets.

- Internal threats and attacks include:

    - Data sharing through public domains or 3rd party people
    - Unauthorized downloading of sensitive data on personal storage drives
    - Unauthorized data transfer using personal cloud storage accounts
    - Social engineering where attackers manipulate employees into giving up confidential data
    - Physical theft of company equipment
    - Abuse of employee privileges to access sensitive company data for personal and malicious motives.

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External Threats and Attacks

- An External Attack Surface, also known as Digital Attack Surface is the sum of an organization's internet-facing assets and the associated attack vectors which can be exploited during an attack. This includes anything from domain names, SSL certificates and protocols to operating systems, servers, IOT devices and network services.

- External threats and attacks may include:
  - phishing
  - malware
  - DDoS attack
  - Malvertizing
  - Hacking through security loopholes
  - Ransomware
  - Physical theft of devices that can offer unauthorized user access
  - 3rd-party apps
  - Malicious USB drops.



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Passive and Active Attacks

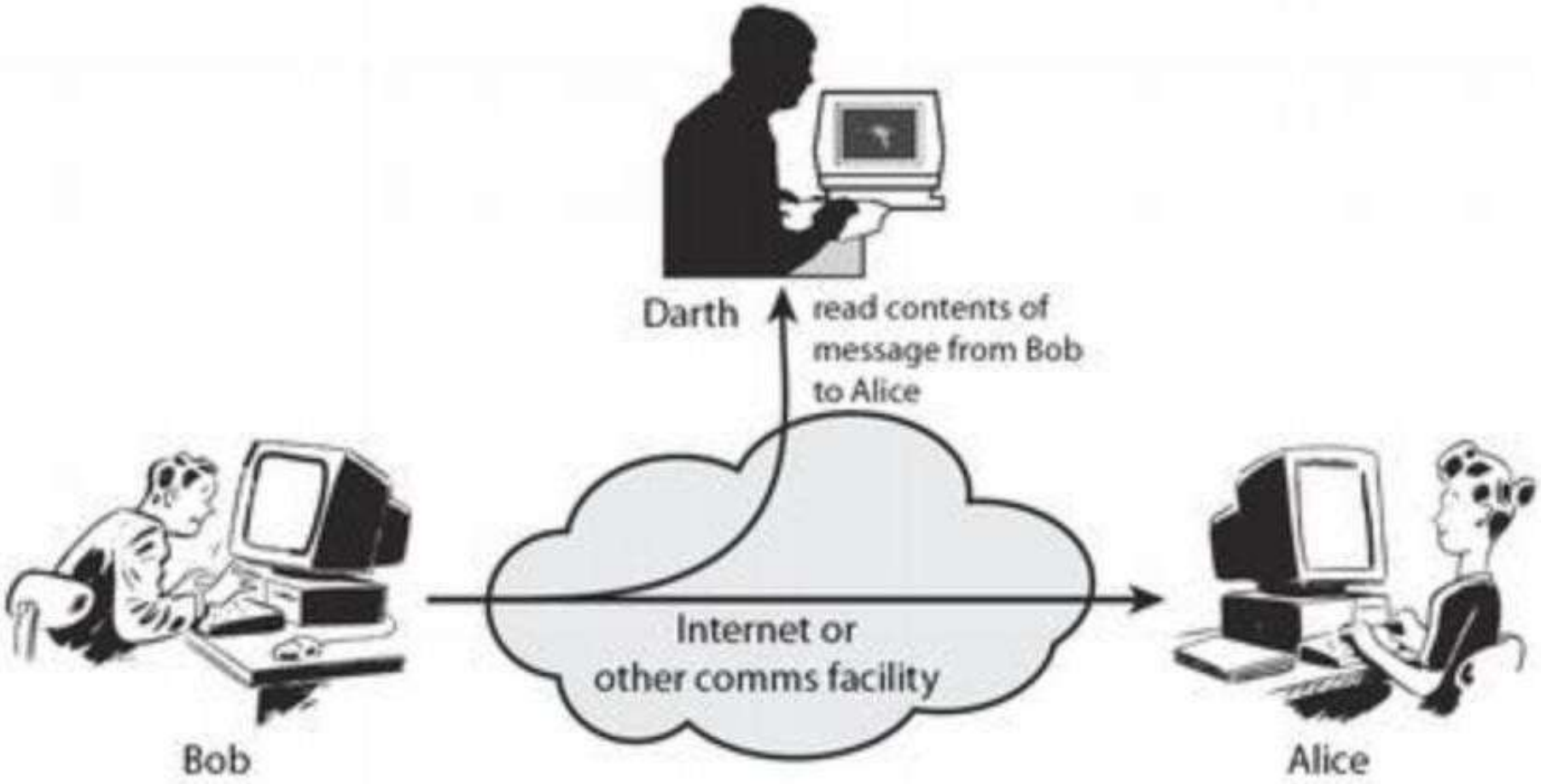
- Active attacks: An Active attack attempts to alter system resources or affect their operations. Active attacks involve some modification of the data stream or the creation of false statements.

- Types of active attacks are as follows:

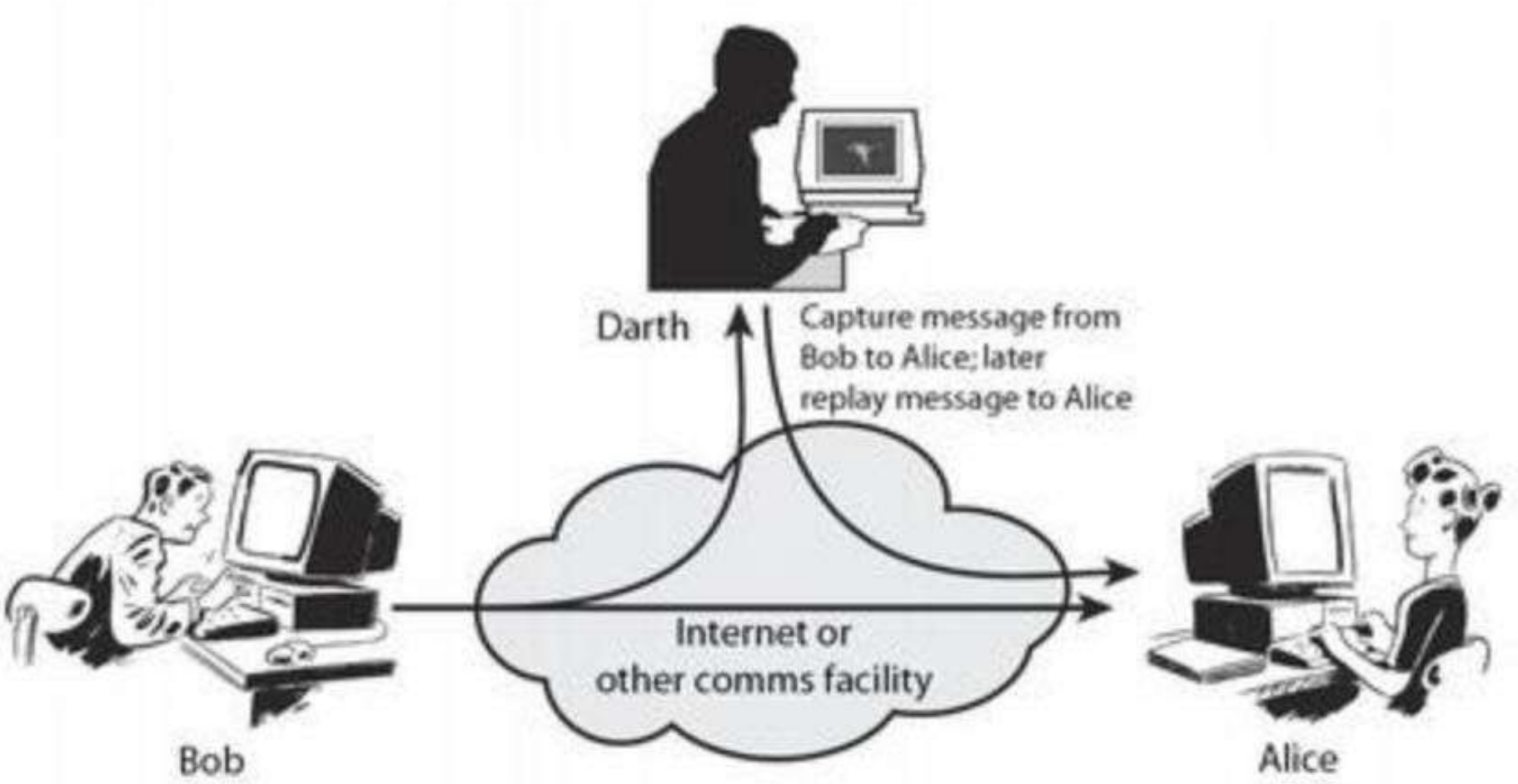
  - Masquerade
  - Modification of messages
  - Repudiation
  - Replay
  - Denial of Service.

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Passive Attacks



Active Attacks



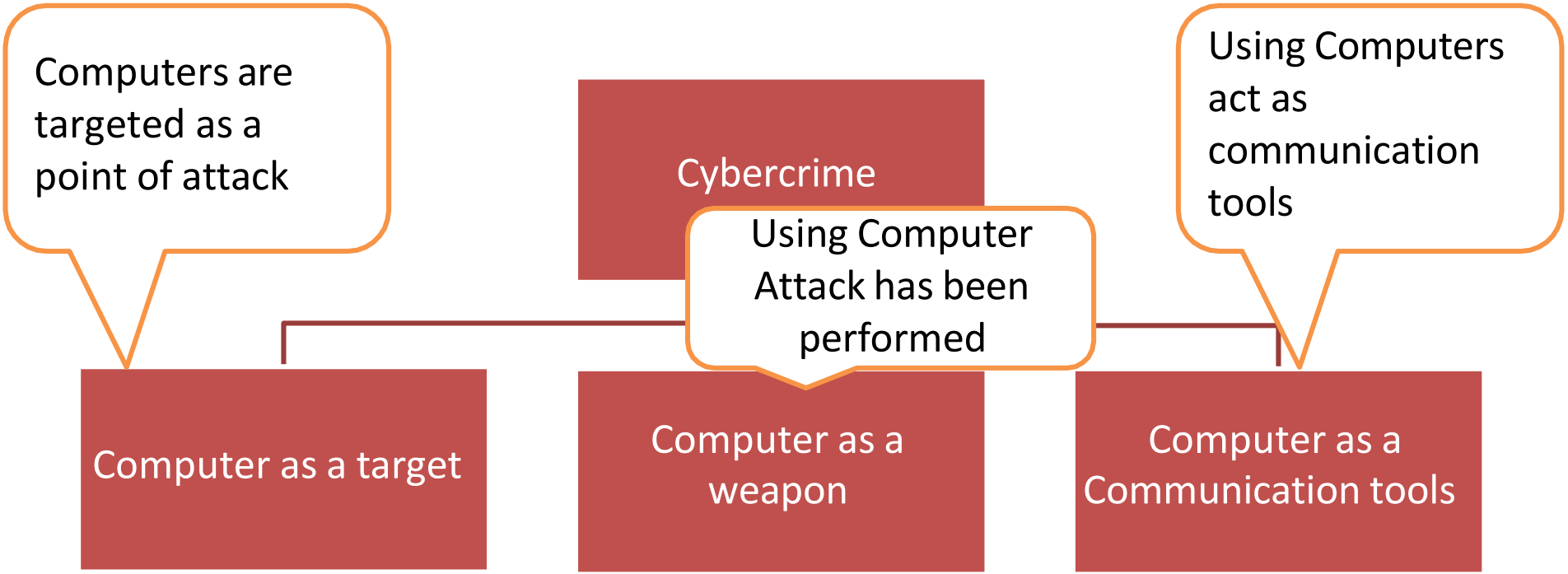
Passive and Active Attacks

- A Passive attack attempts to learn or make use of information from the system but does not affect system resources. Passive Attacks are in the nature of eavesdropping on or monitoring transmission. The goal of the opponent is to obtain information that is being transmitted. Types of Passive attacks are as follows:

    - The release of message content
    - Traffic analysis

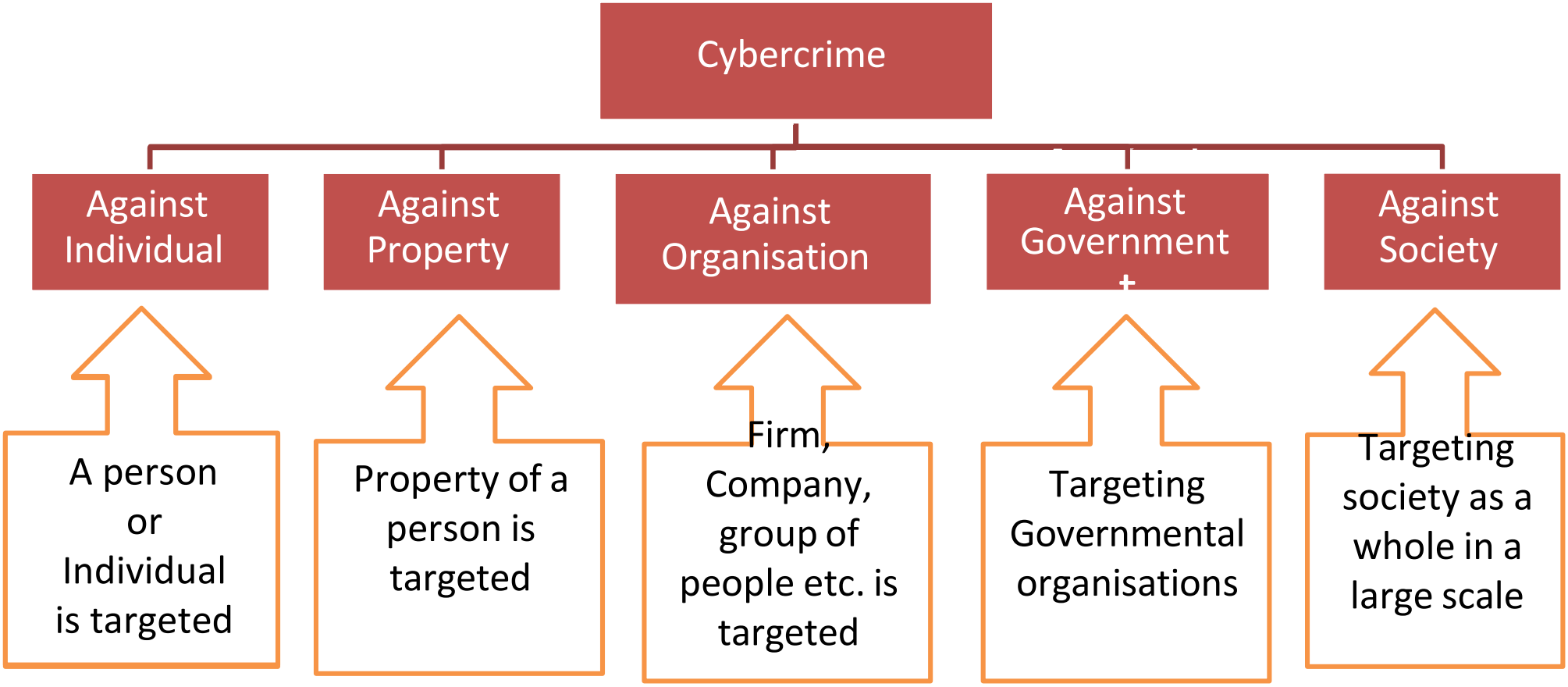
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Computer Roles in Cybercrime



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Categories of Cybercrime



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Module 1:

Lecture 3: Cybercriminals and types of cybercriminals

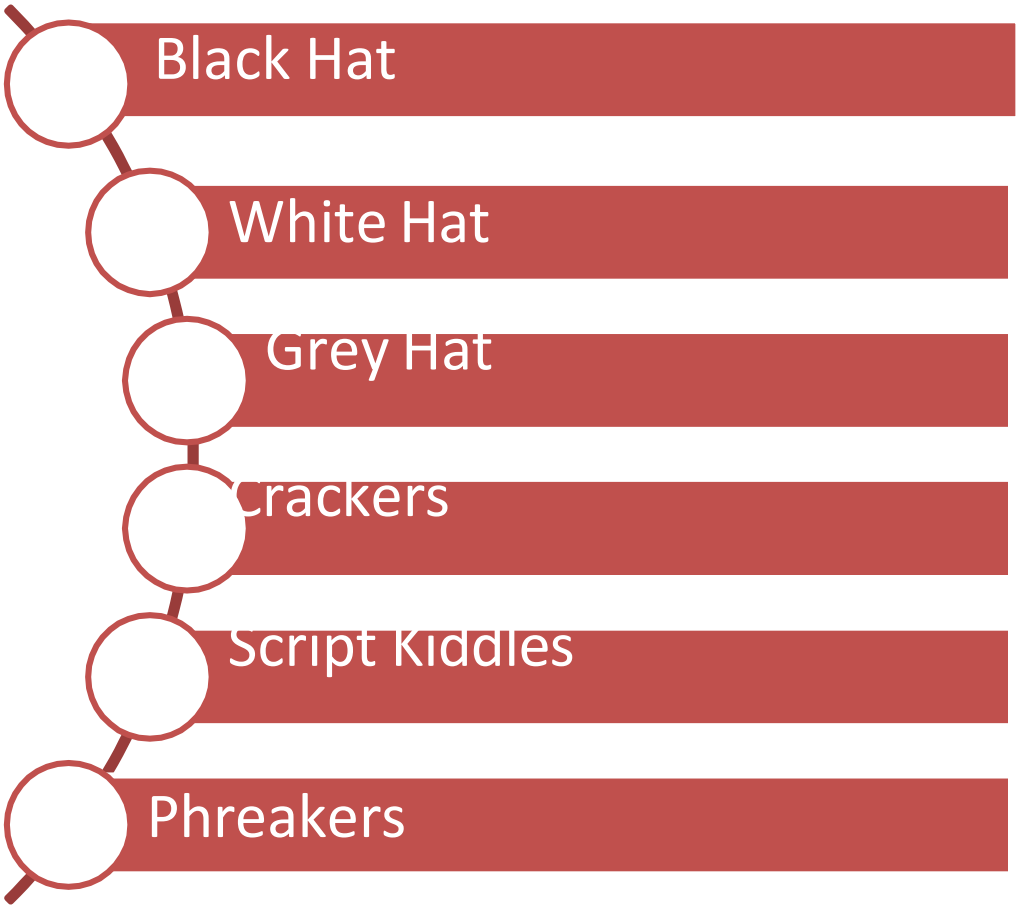


Cybercriminals

- Those who are involved in fraudulent transactions.
- They work in individual or organised group who work in a hierarchical group.
- Organised group create botnet and launch the attacks.
- Cybercriminals are known to access the cybercriminal underground markets found in the deep web to trade malicious goods and services, such as hacking tools and stolen data.

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Types of Cybercriminals



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Types of Cybercriminals

- **Black Hat:** Individuals with extraordinary computing skills, resorting malicious or destructive activities are also known as crackers.
- **White Hat:** Individuals professing hacker skills for defensive purposes and are known as security analysts. They have permission from the system owner.
- **Grey Hat:** Individuals who work both offensively and defensively at various times.
- **Suicide Hackers:** Individuals who aim to bring down critical infrastructure for cause and are not worried for facing jail terms or any other means of punishments.
- **Script kiddies:** An unskilled hacker who compromises system of running scripts, tools, and softwares developed real hackers.

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Types of Cybercriminals

- **Cyber Terrorists:** Individuals with wide range of skills motivated by any religious or political beliefs to create fear by large scale disruption of computer network.
- **State sponsored Hackers:** Individuals appointed by government to penetrate and gain top secret information and to damage information system of other government.
- **Hacktivist:** Individuals who promote a political agenda by hacking, especially by defacing or disabling websites.
- **Crackers:** breaks into the system or violates the integrity of the system with malicious intent.
- **Phreakers:** exploits the secure telecommunication networks. They breaks into the telephone networks illegally, typically to make free long-distance phone calls or tap phone calls.

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Types of Cybercriminals

Hacker	Cracker
Lots of knowledge and Experience	Lots of knowledge and Experience
Good guy	Bad guy
Strong ethics	Poor ethics
No crime	Commits crime
Fights criminals	Is the criminal

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Prevention of Cybercrime

- Always update the computer system regularly with security patched as updates.
- Always select strong password.
- Protect your system using antivirus and firewalls.
- Use encryption services to protect sensitive data.
- Always create backups of important data periodically.
- Secure network (wired/wireless) with strong security protocols and passwords.
- Always ensure that website are secure.
- Never reply to the emails that ask you to verify your information.

https://www.youtube.com/watch?v=mhR0gigfBbA

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Need for a Comprehensive Cyber Security Policy

Security Policies:

Security policies are a formal set of rules which is issued by an organization to ensure that the user who are authorized to access company technology and information assets comply with rules and guidelines related to the security of information.

A security policy also considered to be a "living document" which means that the document is never finished, but it is continuously updated as requirements of the technology and employee changes.

We use security policies to manage our network security. Most types of security policies are automatically created during the installation. We can also customize policies to suit our specific environment.

Need of Security policies

Need of Security policies

1) It increases efficiency.
2) It upholds discipline and accountability
3) It can make or break a business deal
4) It helps to educate employees on security literacy

Common Policy Types

1. Acceptable Use Policy

Defines what constitutes appropriate use of organizational IT resources.

2. Account Management Policy

Sets rules for creating, managing, and deactivating user accounts.

3. Anti-Virus Policy

Mandates the use of antivirus tools, auto-updates, and malware prevention strategies.

4. E-Commerce Policy

Establishes security controls for online transactions and e-commerce platforms

Need for a Comprehensive Cyber Security Policy

- Cybersecurity plays a crucial role in the digital world.
- Securing information and data has become one of the most important challenges in the present day.
- Whenever we expect cybersecurity the primary thing that involves our mind is cyber crimes which are increasing immensely day by day.
- Various Governments and Organizations are taking many measures to stop these cybercrimes.
- Besides various measures, cybersecurity remains a massive concern to several.

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Need for a Comprehensive Cyber Security Policy Contd….

- Companies face potential risks to their systems and data.
- Many cyberattacks take advantage of an organization's employees in some way, exploiting negligence or tricking them into taking action through phishing or social engineering attacks The rise of remote work also poses additional threats due to growing BYOD policies and the potential for faulty equipment to connect therefore to corporate networks.
- Cybersecurity policies help protect an organization from cyber threats and ensure compliance with applicable laws.
- These policies can reduce an organization's risk by training employees to avoid certain activities and enable effective incident response by defining policies for detection, prevention, and remediation.
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Need for a Comprehensive Cyber Security Policy Contd….

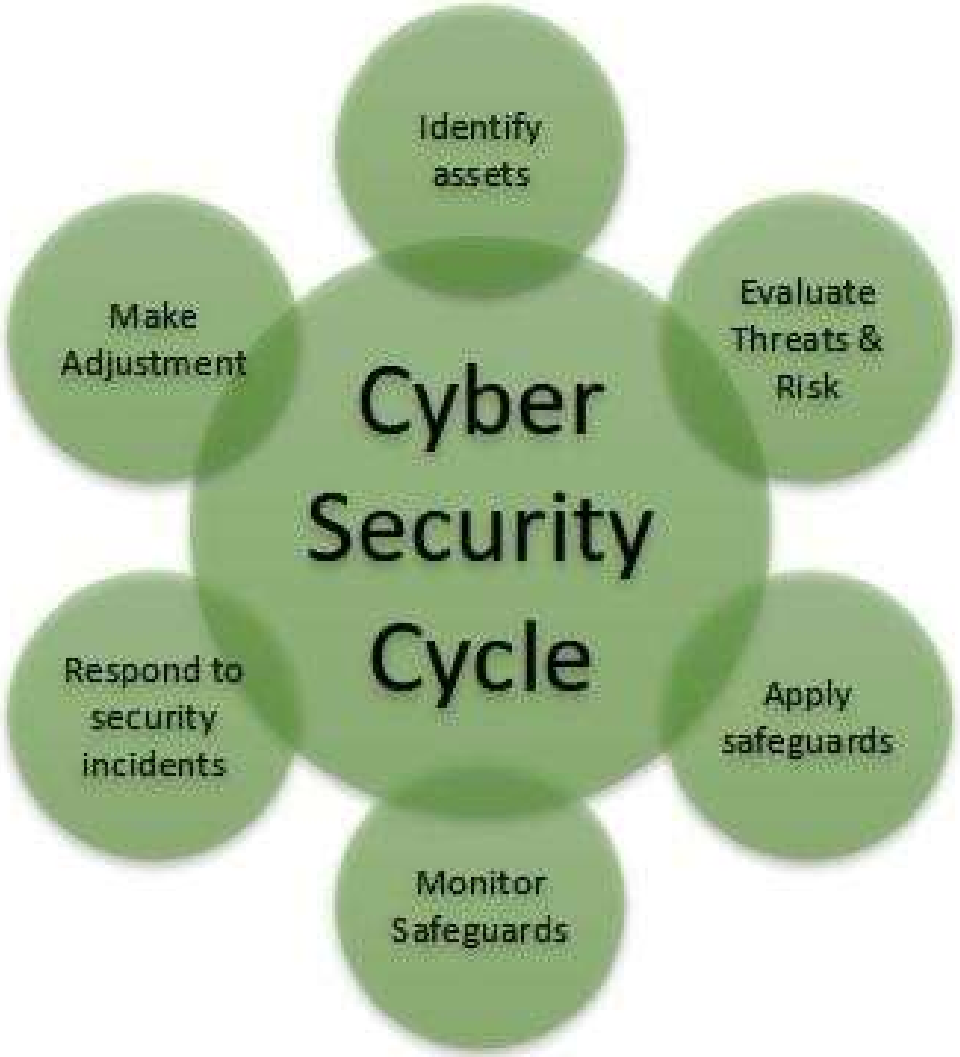


Fig: Cybersecurity Cycle

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Need for a Nodal Authority

- The Indian Computer Emergency Response Team (CERT-In or ICERT) is an office within the Ministry of Electronics and Information Technology of the Government of India. It is the nodal agency to deal with cyber security incidents. It strengthens security-related defence of the Indian Internet domain.
- Under the IT Act, 2000, Section 62, the Central Government of India created the Cyber Regulations Appellate Tribunal (CRAT) as a chief governing body and authority for fact-finding, receiving cyber evidence, and examining witnesses.
- Cybersecurity and Infrastructure Security Agency, or CISA, as it is commonly known. American government has made cybersecurity a top priority.

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Need for a Nodal Authority Contd…….

- CERT-In perform the following functions in the area of cyber security
- Collection, Analysis and dissemination of information on cyber incidents.
- Forecast and alerts of cyber security incidents.
- Emergency measures for handling cyber security incidents.
- Coordination of cyber incident response activities.
- Issue guidelines, advisories, vulnerability notes and whitepapers related to information security practices, procedures, prevention, response and reporting of cyber incidents.

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Need for a Nodal Authority Contd…….

- The Ministry for Electronics and Information Technology (Meity) deals with policy relating to IT, electronics and the internet, including cyber laws. It set up the CERT-In as a nodal agency for co-ordination and handling of cyber incident response activities.
- National Security Agency (NSA) Cybersecurity prevents and eradicates threats to U.S. national security systems with a focus on the Defense Industrial Base and the improvement of U.S. weapons' security. It also strives to promote cybersecurity education, research, and career-building.
- CYBER Stands For: Changing Yesterdays Behavior For Enhanced Results.

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Need for an International convention on Cyberspace.

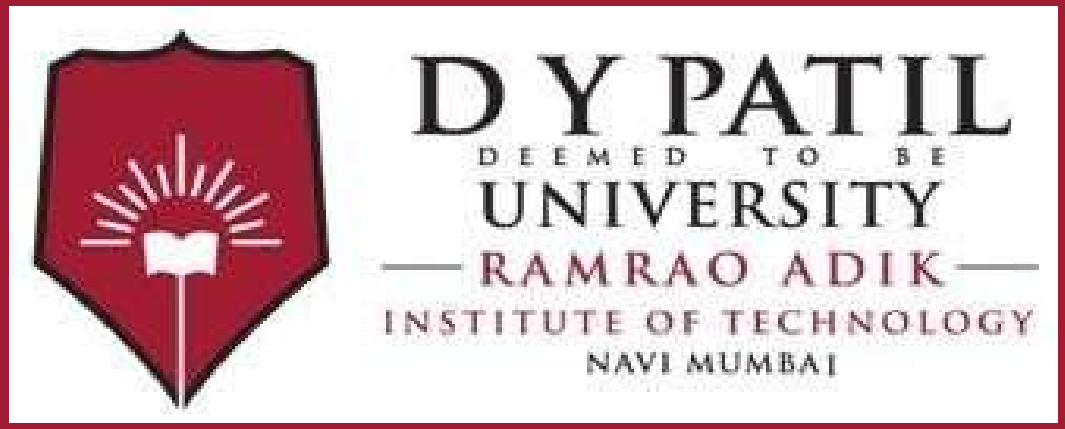
- Cyberspace is a complex environment consisting of interactions between people, software, and services, supported by the worldwide distribution of information and communication technology (ICT) devices and networks.
- An international convention on cyberspace may protect peace, security, and justice in cyberspace, prevent conflicts and maintain focus on cooperation among all nations. A global framework may also develop common legal norms and standards.

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Need for an International convention on Cyberspace.

- A global instrument for cyberspace will be a major step toward building trust, safeguarding information infrastructure, and promoting an open information society at the global level.
- A Geneva Convention or Declaration for Cyberspace may be an initiative by the United Nations institutions in Geneva, including the International Telecommunication Union (ITU), and could be adopted by States at a Ministerial Summit in Geneva.

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Thank You